

Supplementary Information on Soil Dataset Design

Kindly provided by Jorge Locatelli (Colorado State University)

In the SSURGO dataset, some variables can be considered static (i.e., not changing much over time; e.g., texture) and others dynamic (e.g., soil organic carbon and fertility indicators). Because the survey was conducted at different times, the dynamic variables may not reflect the current (or future) status of the field. Therefore, using static variables may be more appropriate for our purposes.

In our modeling work, we usually use the following static variables from SSURGO:

- Sand and clay contents (you can also use the soil type description instead of the numeric texture values);

Soil texture has a direct impact on water dynamics (available water content can be derived), as well as on cation exchange capacity and nutrient retention. Nutrient status and water availability can influence crop choice decisions.

- pH (can change with liming; fertilization may also affect it, but typically not drastically);

This is a good (partial) indicator of soil fertility and the availability of essential elements (e.g., calcium). Nutrient status can influence crop choices.

- Soil depth (restrictive layer depth);

This affects the maximum rooting depth and the soil water storage capacity. Soils with restrictive layers may limit the suitability of crops with higher water demand.

- Drainage classification (there is a separate dataset for tile drainage installation, and you can use them together);

Some crops are more tolerant of poor drainage conditions than others.

- Soil erosion factors (slope) are used by some models and may also be important for your analysis;

Steeper areas are often used for grasses rather than annual crops to reduce erosion risk.

- Crop Productivity Index (this is derived from other static and dynamic variables. I could not find detailed documentation on how it is calculated, but since it incorporates some static variables, it may still be useful);

Areas with higher productivity potential may be used for more profitable crops, which can influence crop choice.

Regarding soil layers, focusing on the top **30 cm** should be sufficient (in addition to using restrictive layer depth), as most crop root systems are concentrated within this layer.